

EDUCATION & SKILLS

M.S. Data Science, Biostatistics Minor — University of Minnesota	Expected May 2027
B.A. Data Science & Sociology — Macalester College	May 2025

Honors

Relevant Coursework: Probability, Statistical Theory, Statistical Modeling, Causal Inference, Advanced Machine Learning, Data Mining

Statistical Skills: Longitudinal data analysis, mixed-effects and multivariate modeling, descriptive statistics, data cleaning and harmonization, reproducible coding in R and Python, IRB and HIPAA compliance, manuscript writing

Programming: R, Python, SAS, SQL, Git, LaTeX; experienced cleaning and analyzing large, messy real-world datasets; Claude, GitHub Copilot

Libraries: R (dplyr, ggplot2, readr, caret, glmnet, randomForest), Python (scikit-learn, NumPy, pandas, TensorFlow/Keras), QGIS, Microsoft Office & G-Suite

EXPERIENCE

Graduate Research Assistant	Spring 2026 — Present
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University of Minnesota Medical Education Outcomes Center, Minneapolis, MN

- Built and maintained a two-section pipeline in R to process two physician survey datasets ($n \approx 16,500$) that used different rural classification systems, producing clean analytic files for presentation.
- Conducted chi-square tests in R on aggregated survey data to evaluate associations of physician satisfaction, rurality, and practice type.
- Prepared manuscript submitted to the *Journal of Rural Health*, including literature synthesis and statistical methods, following author guidelines.
- Followed IRB, HIPAA, and grant documentation requirements throughout the project.

Data Analyst Intern — Climate Team	Fall 2025
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Institute for Agriculture and Trade Policy, Minneapolis, MN

- Cleaned and merged agricultural and satellite datasets from multiple sources (CSV, JSON, and raster formats) in Python and QGIS to prepare them for analysis.
- Built an R Shiny dashboard that lets users explore how crop yields change under different climate scenarios and time horizons.
- Designed a framework for applying ML methods to estimate methane emissions in regions without satellite coverage for future projects.

Data Science Intern	May — Aug 2024
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Securian Financial, St. Paul, MN

- Wrote R scripts and SQL queries to pull, clean, and analyze competitor and market data on a recurring basis from enterprise data systems.
- Built charts, dashboards, and summary reports used by product and strategy teams to make decisions.

Research & Legal Assistant	Sep 2022 — May 2025
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UMN Law School — James H. Binger Center for New Americans, Minneapolis, MN

- Collected and organized case data for legal clinics across the country; Wrote a Python script to visualize patterns in solitary confinement records.

RESEARCH & PROJECTS

Predicting Mild Cognitive Impairment — Wisconsin Longitudinal Study (R)	Fall 2026
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- Trained random forests, SVMs, and neural networks on a longitudinal cohort to predict MCI; used LOOCV and OOB error to evaluate models; handled severe class imbalance through weighting and tuning.

EHR Benchmark Pipeline Reproduction — MIMIC-IV-ED (Python)	Summer 2026
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- Reproduced a published clinical ML benchmark on 448,972 emergency department visits from the MIMIC-IV-ED dataset, implementing logistic regression, random forest, gradient boosting, and MLP models in Python/Keras.

Longitudinal Modeling of Asylum Case Outcomes (R)	2024–25
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- Built a ~ 1.5 M-observation dataset by linking FOIA, USCIS, Census, and IRS sources; resolved duplicate records, inconsistent IDs, and missing demographic data across files.
- Fit hierarchical logistic regression and a 500-tree random forest to model grant outcomes; ran distributional summaries and multicollinearity checks to validate the models.

Causal Inference on Immigration Attitudes in Latin America (R)	Fall 2024
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- Used a difference-in-differences design on 18-country survey data (2018–2023) to estimate how a change in Colombian government affected xenophobic attitudes; ran sensitivity analyses and pre-trend checks to test assumptions.